

MICTURITION REFLEX

- **Trigger / Elicitation** : Elicited by the stimulation of **stretch receptors** located in the wall of the **urinary bladder** and **urethra**.
- **Initiation Pathway** :

300 – 400 mL of urine is collected in the bladder



↑ Intravesical pressure increases



Stretched bladder wall



Stimulation of stretch receptors



Generation of nerve impulse

PATHWAY OF MICTURITION REFLEX

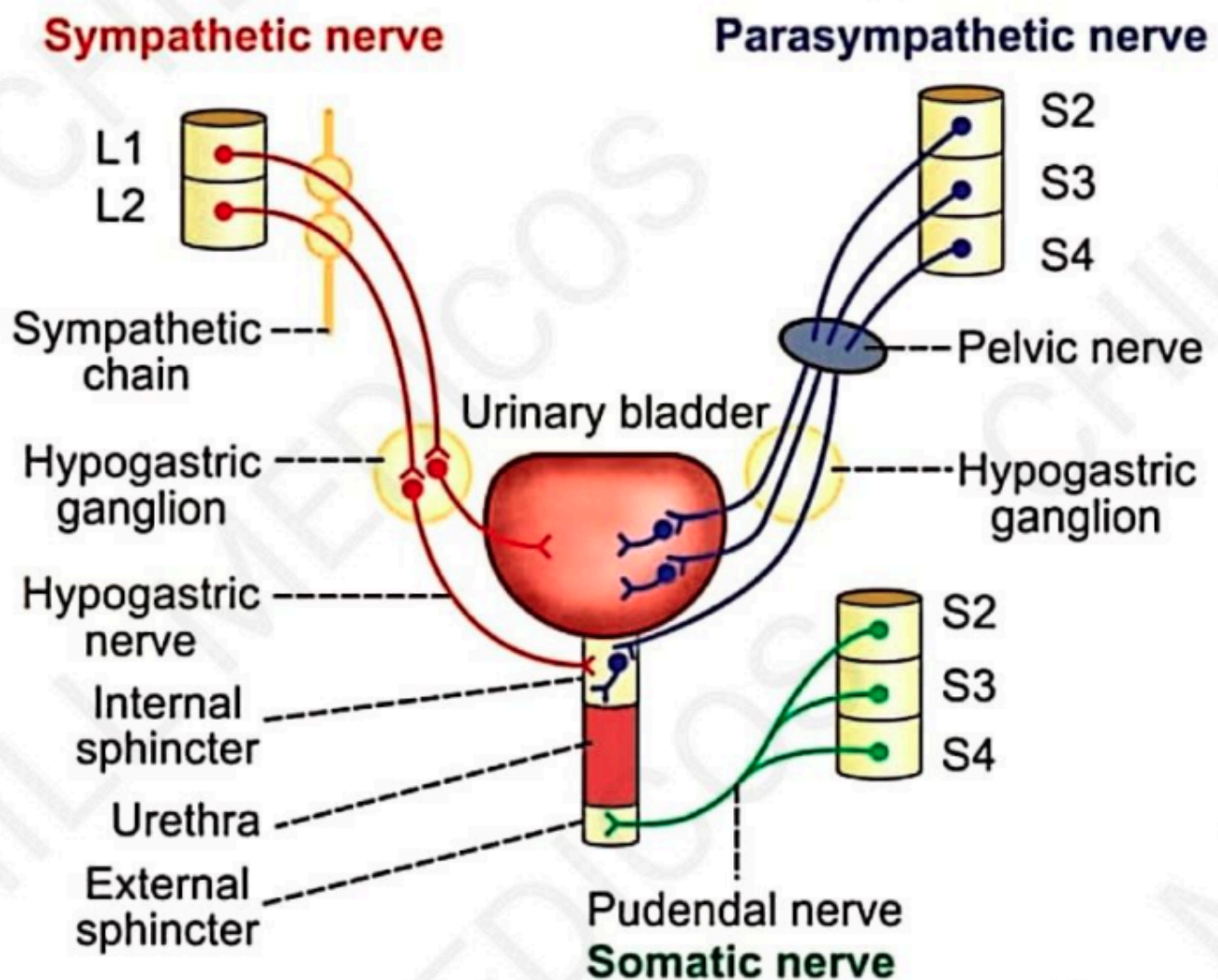


FIGURE 57.3: Nerve supply to urinary bladder and urethra

Filling of urinary bladder



Stimulation of stretch receptors

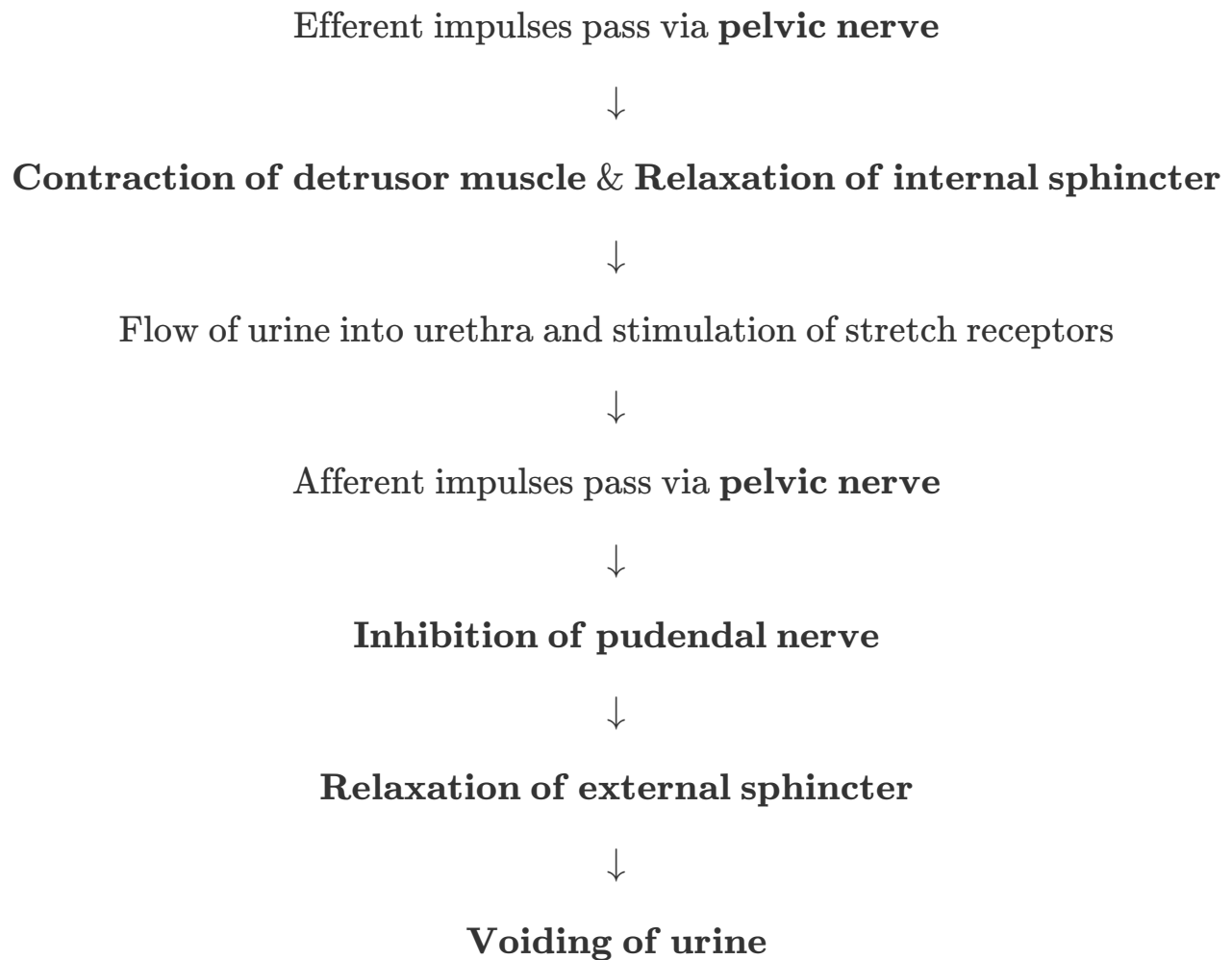


Afferent impulses pass via **pelvic nerve**



Sacral segment of spinal cord





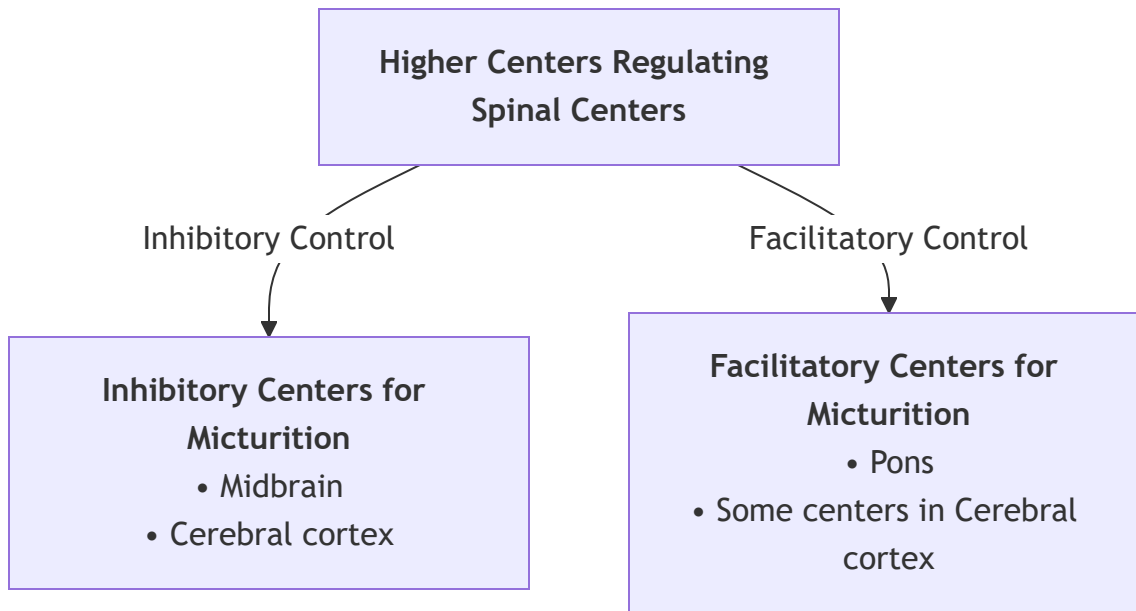
Key Feature: Self-Regenerative Nature

- Once the micturition reflex begins, it is **self-regenerative**.
- Initial contraction of the bladder further activates the stretch receptors.
- This still further ↑ **increases the sensory impulses** from the bladder and urethra.
- The cycle continues until the force of contraction of the bladder reaches its maximum.
- **Complete Voiding** : The urine is voided out completely.
- **Abdominal Pressure** : During micturition, the flow of urine is facilitated by ↑ **increased intra-abdominal pressure**.

HIGHER CENTERS FOR MICTURITION

- **Spinal Centers** :
 - Sacral region

- Lumbar segment
- Spinal centers are **regulated by higher centers**:



APPLIED PHYSIOLOGY

- **1. Atonic Bladder** : Due to **destruction of sensory nerve fibers** (loss of afferent stretch reflex).
- **2. Automatic Bladder** : Characterized by a **hyperactive micturition reflex** (loss of inhibitory control from higher centers).
- **3. Nocturnal Micturition** : **Involuntary voiding of urine** during the night.